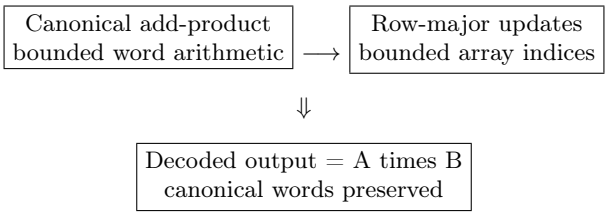
Canonical polynomial product accumulation

T-1209; 2026-11-01. Producer proof and separately attributed review evidence.



Derived result. The native-shaped row-major product model preserves canonical coefficient words and decodes to the product of the two input polynomials. The proof includes the single modular add-product update, each coefficient row, all rows, zero initialization, both empty-input branches and final trimming. Assume prime $0<p<2^{31}$ and canonical input words. The native interpretation additionally requires odd characteristic, extension degree one and diagnostic tables disabled.

Construction. For a row multiplier x at offset i, each bounded array update adds x times the corresponding second-input coefficient. After that row, the decoded accumulator increases by x X^i times the second polynomial. Induction in first-input order gives the full product. Modular multiplication and addition retain canonical words; intermediate multiplication and addition are bounded by p^2 and 2p respectively. The finite-index model proves each selected array index is in range. Trimming preserves the decoded product, including all-zero output.

Checks. Ten main declarations and eight kernel controls passed on their first attempts. Separate axiom readback covers all 18 declarations, using only propext, Classical.choice and Quot.sound. Final native check exits are zero and input bindings unchanged. Controls cover two linear factors, an internal zero coefficient, squaring, left/right empty inputs, trailing zeros, an all-zero product and repeated contributions to shared coefficient slots. These are proof controls, not sampled performance measurements.

Independent evidence. A separate source auditor pinned three_leads.rs and geometry_events.rs, documented the row-major product order and the actual pneg-then-padd subtraction order, and identified caller premises. A separate isolated monic-remainder replay checked 19 main declarations, 16 producer controls, four additional reviewer controls and 39 axiom readbacks. It found the arguments faithful at their stated model scope, while recording that its review was not blind re-derivation or a full clean dependency rebuild. These findings concern the prior monic packets, not an independent product proof verdict. The separate product reviewer recorded a prospective local plan and derived the native invariant before reading the proof bodies. Fresh isolated replay passed all ten main declarations, eight producer controls and six reviewer controls, with 24 axiom readbacks. It found no defect in the scoped model. Precommitted published dispatch, full dependency rebuild, independent kernel/independence-checker gates and compiled equivalence remain outside this review. Detailed findings are in reviews/product.

Scope. The source audit identifies native vector allocation, usize arithmetic, quotient writes and checked instrumentation counters as compiled-execution obligations outside the polynomial-value model. For Jacobian.valid, the participating ordinate prefix has length at most two, but the theorem itself permits arbitrary model lists. Canonicality, field dispatch and $p<2^{31}$ must be established by callers; Jacobian.valid does not check all of them. Native multiplication decoding is now modeled; native pneg, padded padd and the composed f-v^2 residual remain next. No production Rust changed, no scientific worker launched, and no runtime/cost ratio is claimed.

Sources. Native src/cryptanalysis/three_leads.rs pmul at lines 835-847; the independent residual source audit; pinned Mathlib Polynomial/Coeff.lean; existing canonical UInt64 multiplication/addition, polynomial coefficient traversal and trim proofs. This is a source-shaped implementation of ordinary polynomial convolution, with no novelty claim. Existing imported compiled dependencies retain their disclosed trust boundary.